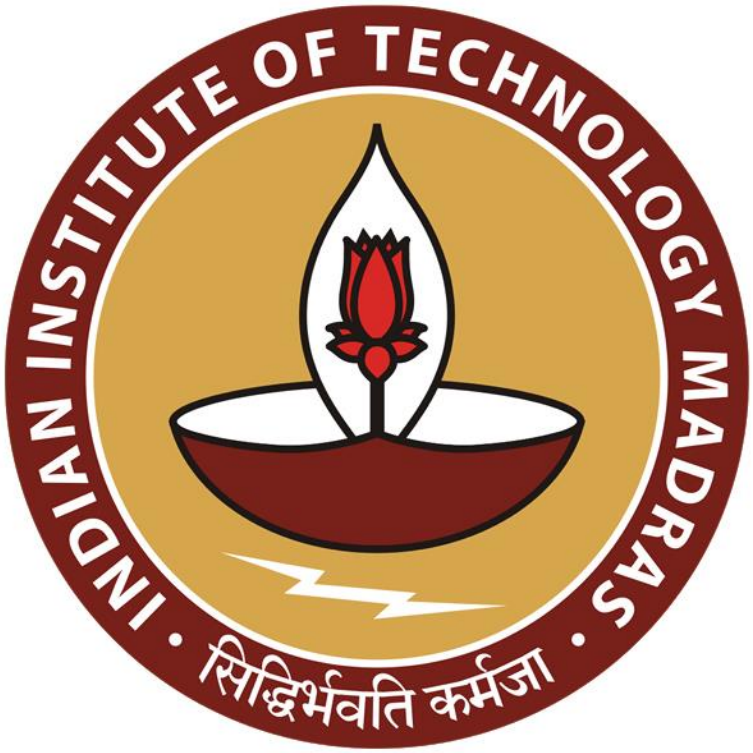


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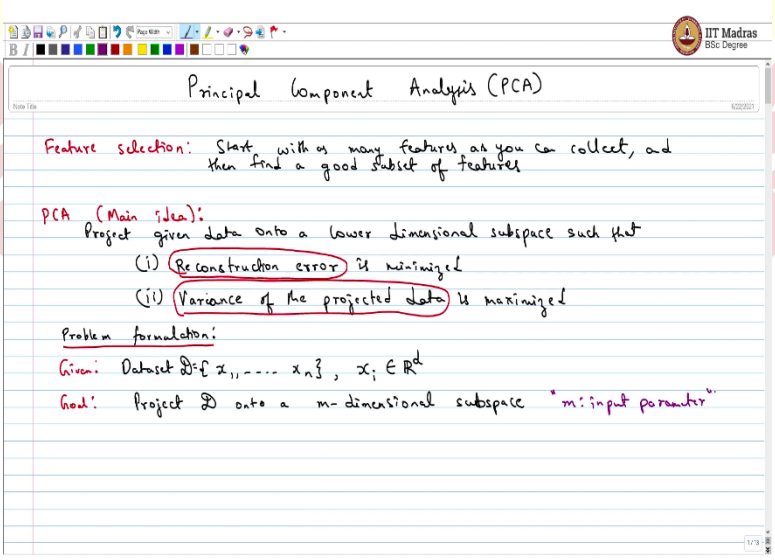
Professor. Prashanth L A
Department of Computer Science & Engineering
Indian Institute of Technology, Madras
Principal Component Analysis

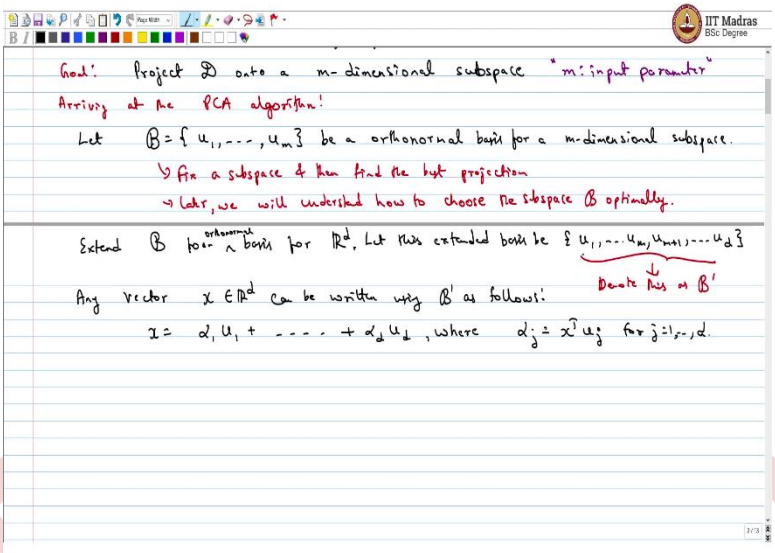
Welcome to lecture 3, in week 6 of machine learning foundations. This is the last lecture of linear algebra, where we will look at a very important topic, principal component analysis, which is a dimension, dimensionality reduction method. Now, we will start in this PCA has two viewpoints, you could view it as either minimising the so-called reconstruction error, which we will define concretely. Or you could also see it as maximising variance of the data. And we will start with the first viewpoint.

In this lecture, we will define what the reconstruction error is for a given set of data points, and then try to understand what PCA does. And we will look at an example subsequently. And this is a two-part lecture simply because PCA discussion will take longer than one video. So, we will continue this discussion in the next lecture also, and then look at an example to see how PCA is minimising the reconstruction error.

Subsequently, we will look at the second viewpoint that is of PCA as maximising variance and show that how these two viewpoints are equivalent. You could view PCA has either minimising the reconstruction error or maximising variance. So, we will discuss that in the subsequent lectures.

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Principal component analysis is a method that is used for feature selection in the context of machine learning problem. And the idea of machine's feature selection to put it in very simple words is, you start with as many features as you can collect, and this just means that you are going to have high dimensional features.

And then what you do is, you use some methods for dimensionality reduction, PCA is a very popular alternative. So, what you do is then find a good subset of features. This is the broad problem of feature selection. And what does PCA, main idea? The main idea behind PCA is project given data onto a lower dimensional subspace such that it, a PCA has two interpretations and both are valid in the sense one, and some project onto a low dimensional space such that reconstruction error, we will make all of this precise.

Reconstruction error is minimised or alternatively variance of the projected data is maximised. So, what PCA does is, you can derive the PCA algorithm by looking at variance of projected data, projected data is maximized. So, both are valid, so we will first look at PCA, the main idea by understanding how it minimised the reconstruction error. For this, here is a problem formulation.

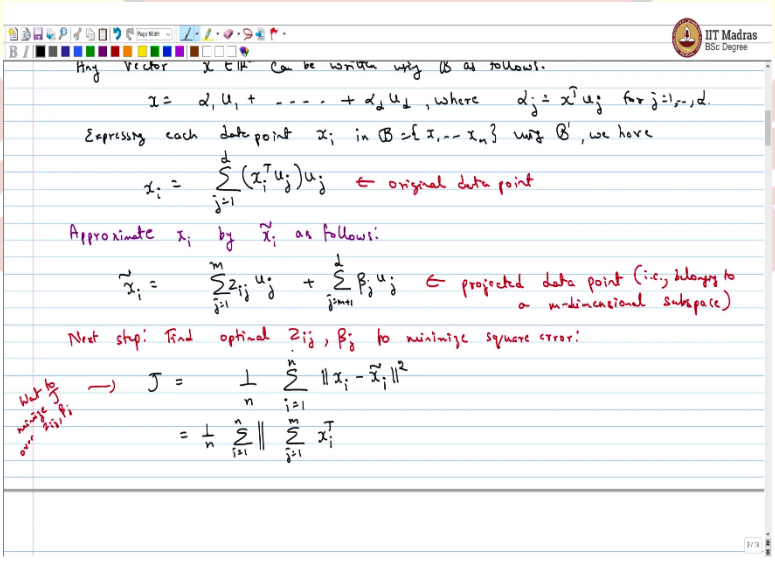
Suppose, you are given a data set which is D= $\{x_{1,\ldots\ldots\ldots}, x_n\}$, where each x_i belongs to the d dimensional Euclidean space. Now this is given, this is given. So, what is the goal? Project, let me call this data D, project D onto a m-dimensional subspace. Now, important m, is an input parameter, you fix this parameter beforehand, imagine you fix a lower dimensional, you fix the dimension of the space, subspace that you want to project onto.

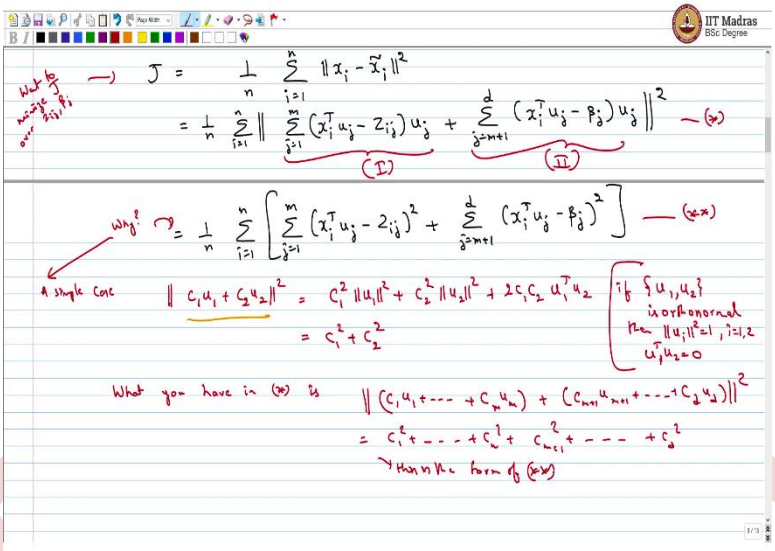
And now, given this parameter, you want to find the best m dimensional subspace to house this data. So, we will arrive at a PCA algorithm by looking at I mean in a constructive manner, I will not specify the algorithm and try to deconstruct. Let us start with the problem itself. So, given this goal, so, let me say arriving at PCA algorithm, or the build up to the PCA algorithm.

So, let B = $\{u_1,\ldots\ldots\ldots,u_m\}$ be a orthonormal basis for a m dimensional subspace. Let us understand this. So, the way to understand this is fix a subspace and then find the best projection. That is what we do, that is what we do, later we will understand how to choose the subspace optimally. So, there are two things to try for the sake of understanding, imagine you are given m vectors in some arbitrarily chosen m dimensional subspace.

And now, given this m vectors, you want to now find the best projection of the data. So, let us do this. So, now, what we can do is you can extend B to the basis for the original D dimensional subspace that houses the data. So, let this extended basis B, you have $\{u_1,\ldots\ldots\ldots\ldots,u_m\}$ and then you add the remaining d - m vector. So, $\{u_1,\ldots\ldots\ldots\ldots,u_m, u_{m+1},\ldots\ldots\ldots\ldots\ldots u_d\}$. So, this is a basis sent to an orthonormal basis and worked on normal basis. Now, given this basis, denote this as B prime. Now, this is a full basis for the D dimensional space. So, any vector $x\epsilon\mathbb{R}^d$ can be written using B prime as follows. Now, how do we write this, $\alpha_1 u_1 + \cdots . + \alpha_d u_d$ where, how do you figure this coefficients or the scalars, where $\alpha_j = x^T u_j$ for j= 1,2,…. d. This is fairly straightforward. This is linear algebra, 1 0 1 given a basis, orthonormal basis.

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How can I express any vector in a linear form? Now, so, expressing each data point x_i in the set B that we have $\{x_{1,\ldots\ldots}, x_n\}$ using B prime, we have $x_i = \sum_{j=1}^{d}(x_i^T u_j)u_j$ .Now this is the original x_i, this is original data point. How do we project this onto the m dimensional space?

So, let us want to approximate x_i by x_i tilde as follows, so how do you do this, what you do is, let us define xi tilde $= \sum_{j=1}^{m} z_{ij}u_j + \sum_{j=m+1}^{d} \beta_j u_j$. Now this is the projected data point, the point that belongs to the m dimensional subspace. Now we need to figure the coefficients. Now, next step, find optimal z_{ij} and β_j. I will just leave it. So, how do you find this, for this we need to minimise certain square error. So, let me define this. So, the error that you want to minimise is $J = \frac{1}{n}\sum_{i=1}^{n}||x_i - \bar{x}||^2$, therefore want to minimize j over, over this coefficients zij, β j. This is what you want to do.

Now, before we go on to minimising this, I will just rewrite this objective now by expanding the square, this is nothing but $J = \frac{1}{n}\sum_{i=1}^{n}|| \sum_{j=1}^{m} x_i^T u_j - z + \sum_{j=m+1}^{d}(x_i^T u_j - \beta_j)u_j||2$

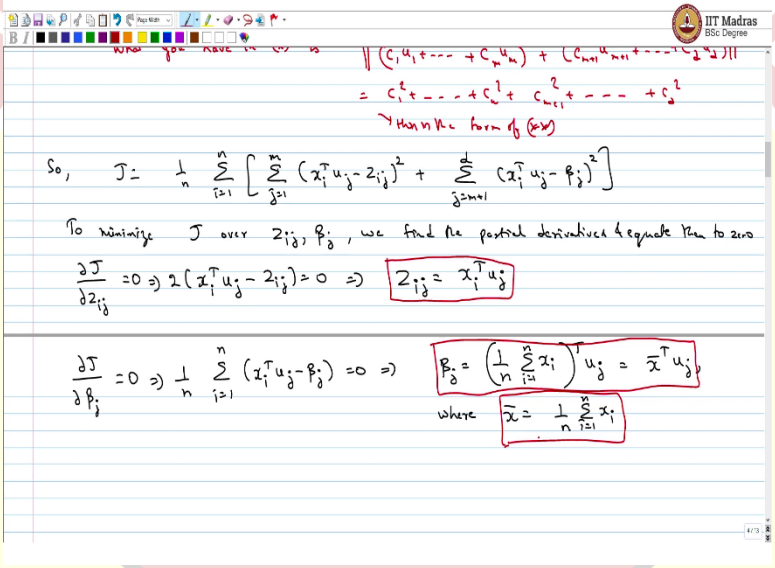
Now, this is what you have in the norm, but what is important is this simplifies if you expand the norm. So, what happens is this is like, if you expand this, so, recall the L2 norm, I hope you know what the L2 norm is. So, if I strike out the mean conclusion then try to argue this is really the case. So, the question is why is this true, why?

A simple case, imagine you have norm of say $||c1u1 + c2u2||^2$, this is going to be $c_1^2||u_1||^2 + c_2^2||u_2||^2 + 2\, c_1c_2\, u_1^T\, u_2$. Now, if $\{u_1, u_2\}$ is orthormal, then $||u_i||=1$, i = 1 ,2 and $u_1^Tu_2 =0$. Now, this just means that this is equal to $c_1^2 + c_2^2$. And the same thing is happening here.

Now, if you look at this let me call this term 1, and call this term 2, then what you have in star is $\|(c_1u_1 + \ldots\ldots c_mu_m) + (c_{m+1}u_{m+1} + \ldots\ldots\ldots c_d u_d)\|^2$.

Now, this is going to be equal to just the individual square simply because when I take, I mean I can extend this logic here, I can extend this logic all the way here. So, this I would get is $c_1^2 + \ldots\ldots c_m^2 + c_{m+1}^2 + \ldots\ldots\ldots c_d^2$, which is, let me call this double star and this is the form of double star. Simply because you have orthonormal basis the cross terms vanish and for each of the individual term since, you i's are all unit norm, you get the individual squares. So, we have, so, let us get back to PCA.

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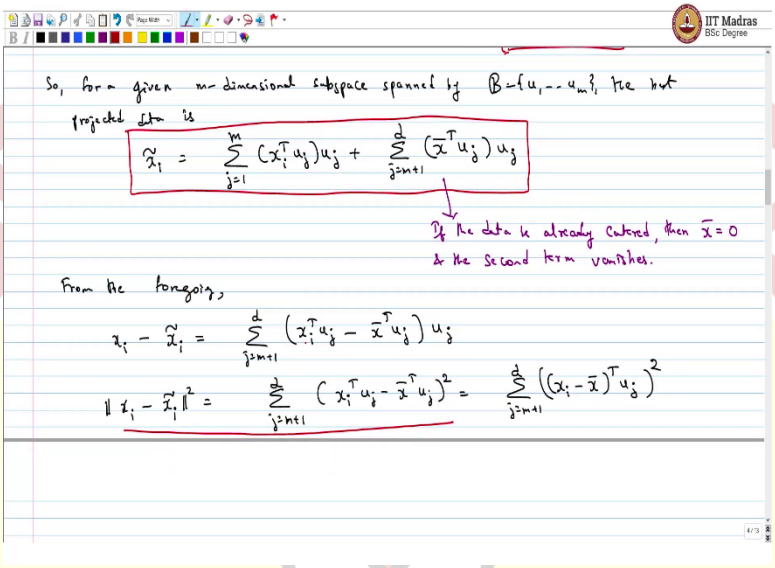
So, what we have is but $J = \frac{1}{n}\sum_{i=1}^{n}[\sum_{j=1}^{m}(x_i^T u_j - z_{ij})^2 + \sum_{j=m+1}^{d}[(x_i^T u_j -]\beta_j)^2$. Now, to minimise J over z so, we need still need to figure what are the best coefficients z_{ij} and βj, we find the partial derivatives and equate them to 0.

So, what are the partial derivatives? Let us do $\frac{\partial j}{\partial z_{ij}}$ if you see z_{ij} appears only in one time in this entire. And now if you take the derivative of that, you just get $2(x_i^T u_j - z_{ij})= 0$ which just implies the most important thing $z_{ij} = x_i^T u_j$. So, we got a first version.

Now proceeding similarly, you want to do something to β. So let us $\frac{\partial j}{\partial \beta_j}$. Now, I have to equate this to 0. So, this β j appears within a summation, it is not just one term, it is a summation. So, what I would get is 1 by n summation i equals 1 to n, let me write it this implies this.

So, I am going to equate this $\frac{1}{n}\sum_{i=1}^{n}(x_i^T u_j - \beta_j)$ this is what will be equal to 0, this just implies if I rewrite this β_j $(\frac{1}{n}\sum_{i=1}^{n} x_i)^T$ u_j this is what you would get. I will denote this as $\bar{x}^T$ uj, where $\bar{x}$ is the mean of the data which is $\frac{1}{n}\sum_{i=1}^{n} x_i$, so this is the expression β_j is this, we take the average point and take the dot product with u_j so the average point is denoted here.

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So, clubbing this all together substituting back in x_j tilde, so, for a given m dimensional subspace spanned by $\{u_1$ ……to $u_n\}$ the best projected data is given by xi tilde $=\frac{1}{n}\sum_{j=1}^{m}(x_i^T u_j)u_j + \sum_{j=m+1}^{d}(\bar{x}_i^T u_j)u_j$.

Just a minor note if the data is already centred, then $\bar{x} = 0$ and the second term vanishes. The data is already centred, which is what usually people do when you do PCA. And you can take $\bar{x}$ out of the equation and then what you are left with is just the first term here, and it is clear that it is an m dimensional space.

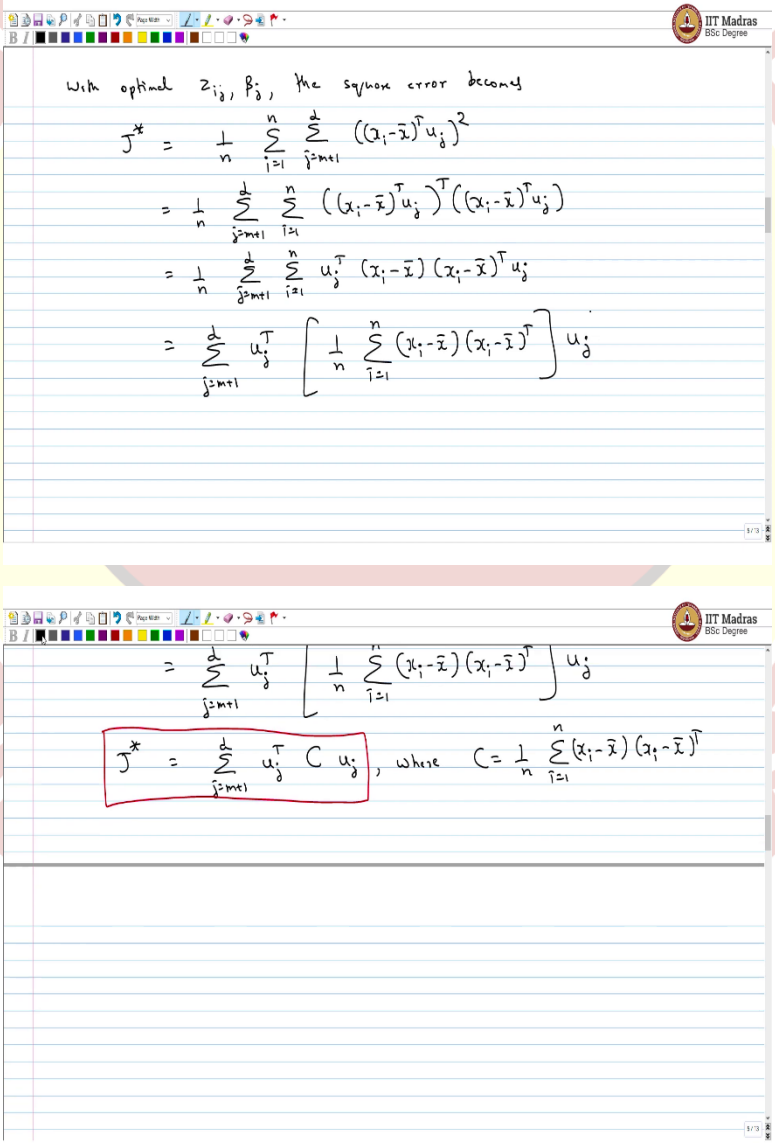
Now, let us proceed and figure which is the best but which is the, which has the best use to pick. But before that, let us understand so, from the foregoing, what you have is x_i - x_i tilde equals, if you look at x_i - x_i tilde, the first m terms coincide just for up for x_i it is each is $x_i^T u_j$.

Now up to m, if z_{ij} is the same as $x_i^T u_j$ up to m, then the first m terms cancel when I take the difference from x_i to x_i tilde, and we, I will be left with the d - m terms. So, x_i - x_i tilde is nothing but $\sum_{j=m+1}^{d}(x_i^T u_j - \bar{x}_i^T u_j)u_j$.

Now, using the logic that I have said before, again, when I take the normal square, then what I would get is individual md squares, which is $\sum_{j=m+1}^{d}\left(x_i^T u_j - \bar{x}_i^T u_j\right)^2$. This is the same as $\sum_{j=m+1}^{d}\left((x_i - \bar{x})^T u_j\right)^2$.

Now this, for this equality follows by the logic that I am explained here, the same as this when you take the square you get the individual coefficients squares, and the individual coefficients are $x_i^T u_j - \bar{x}_i^T u_j$. So, it will be sum of squares of this coefficients, simply because again, you have orthonormal set um + 1 until ud is still a orthonormal set.

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So, what we have is with optimal z_{ij}, β_j the square error becomes what you have is, let me call this J^*, the optimal because it is optimal for a given by $\frac{1}{n}\sum_{i=1}^{n}\ \sum_{j=m+1}^{d}\left((x_{\ i}-\bar{x})^T u_j\right)^2$. This is what we have, we are going to further simplify this.

Now if I continue simplifying this, then what I would get is $\frac{1}{n}$, I am going to interchange the summation, so let j equals m + 1 to d summation i equals 1 to n. And then this square can be written as dot product with itself, which is writing x_i - $\bar{x}^T$ uj, this product let us say xi - $\bar{x}^T$ uj .

$$\frac{1}{n}\sum_{j=m+1}^{d}\ \sum_{i=1}^{n}\left((x_{\ i}-\bar{x})^T u_j\right)^T\left((x_{\ i}-\bar{x})^T u_j\right).$$

Now this again, I can write this as summation j equals m + 1 to d, i equals 1 to n. So, the reason I am doing all of this is I want to extract something with just xi and $\bar{x}$, I want to extract a matrix with just xi and $\bar{x}$ outside and keep the u_j much simplified form, so that I can figure how to find this u_1 to u_n and so that certain the projected error is minimised. So, to do that, let us continue this calculation.

So, what I have is this transpose basic rule is u_j^T $(x_i - \bar{x})$ $(x_i - \bar{x}^T u_j)$. Now this I can write as summation j equals m + 1 to d, and I am going to bring the u_j outside because it is in a summation with variable i here. So, so the u_j can come outside, this can come outside, the other u_j also can come outside, I am just left with the rest.

So, u_j^T and then I am going to write this as 1 by n summation i equals 1 to n this is a very important matrix that will come up now, x_i - $\bar{x}^T$ whole thing into u_j. So, I am going to call this matrix C, summation j equals m + 1 to d u_j^T C u_j, where C is this matrix, which is equal to $\frac{1}{n}\sum_{i=1}^{n}(x_{\ i}-\bar{x})(x_{\ i}-\bar{x})^T$ So, this is what we have as J^* and now you need to find the best use, using this formulation. So, we want to focus on this form and try to figure what is the best way. So, let us consider a very simple case. So, the summary of today's lecture, in today's lecture, we started a discussion of PCA and we are going with the PCA as minimising the reconstruction error viewpoint in this lecture.

So, for a given set of data points in some D dimensional space and an input parameter which is necessary for PCA which gives the dimension of the space where you want to project the data onto given this parameter say m. We are trying to understand what is the best way to project the data on to this lower dimensional space, with this input parameter, so that the reconstruction error is minimised.

So, we defined what reconstruction error is and started the discussion about how we can go about finding the best lower dimensional space. So, that the reconstruction error is minimised. Well, this is a two-part lecture, we will continue this discussion in the next video also.

And then, we will be able to arrive at the final PCA algorithm, which gives you the recipe for finding this best m dimensional subspace onto which you can project the data on to, so that reconstruction error is minimised. In the next lecture we will also after giving the recipe for PCA, we will look at a very simple example to illustrate the ideas of the PCA algorithm.

